

## Tutorial 1

1. Give comparison between mobile ad hoc network (MANET) and wireless sensor network (WSN)

2. In wireless transmission, the receive power  $P_R$  can be expressed as the function of transmit power  $P_T$ , distance between transmitter and receiver  $D$ , and path loss exponent  $E$  as follows:  $P_R = P_T/(D^E)$ . Consider the two-hop wireless transmission where node A is the source, node B is the relay, and node C is the destination as shown in the figure below. Node B is located in between node A and node C. In the first time slot, node A transmits the packet to node B. In the second time slot, node B transmits the packet that node B receives from node A in the first time slot to node C. To receive the packet from node A successfully, the receive power at node B must be at least  $P_{RB}$ . Similarly, to receive the packet from node B successfully, the receive power at node C must be at least  $P_{RC}$ . Derive the optimal location of node B such that the sum of transmit power used by node A and node B is minimized. Note that the transmit power used by node A and node B at the optimal location must ensure that node B and node C can successfully receive the packet.

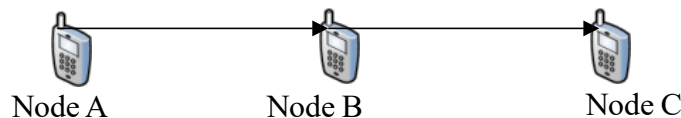


Figure for Question 2

3. Consider the service area as shown in Figure Q1. The base station located at point A transmits data to a user with the transmit power of 15dB.

- If the receive power of the user is larger than or equal to 15 watts, the user can receive the data at the rate of 10Mbps.
- If the receive power is larger than or equal to **5 watts**, the user can receive the data at the rate of 5Mbps.
- If the receive power is larger than or equal to 1 watt, the user can receive the data at the rate of 1Mbps.
- If the receive power is less than 1 watt, the user cannot receive data at all (i.e., the data rate is zero).

The user is at point B which is located 500 meters from point A for 50% of the time. For another 50% of the time, the user can move and be at any point in the service area uniformly. The receive power in watt is given by  $P_R = P_T/(D^E)$ , where  $P_T$  is the transmit power in watt,  $D$  is the distance between the base station and the user in kilometer, and  $E=3$ . Calculate the average data rate of the user.

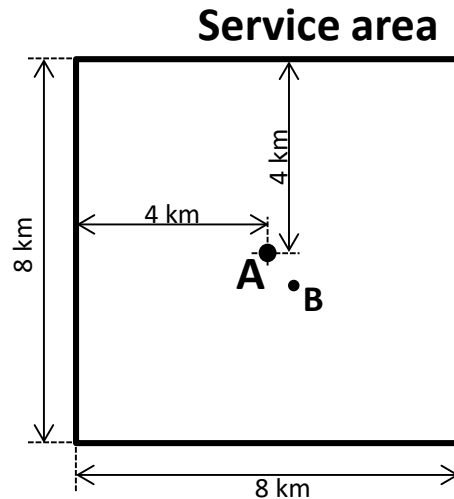


Figure for Question 3

4. A centralized wireless network uses CDMA for medium access. Two nodes X and Y are assigned the following code sequence:

X:  $\langle -1, -1, 1, 1, -1, 1 \rangle$

Y:  $\langle -1, 1, 1, 1, 1, -1 \rangle$

- a. Determine if the codes of X and Y satisfy the orthogonality property.
- b. If X and Y simultaneously transmit the bits 1, show how the receiver can receive correctly each transmission.
- c. Why is it necessary to define a threshold for CDMA reception?

5. Consider a co-located network of 10 nodes using the ALOHA protocol. The raw bandwidth of the radio channel is 10 Mbps. The packet length is fixed at 1000 bits. Packet arrival is random, with each node generating 400 packets every second on the average.

- a. Calculate the throughput of the network.
- b. Also calculate what percentage of the packets encounter collision.

6. Repeat the above calculations for slotted ALOHA.